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WO-400-04 — Remake flanged joint 3"-P300-P-1015/B12 on 3"-P300-P-1015

Area	A400 Pumps	**Task**	GJ Flanged joint remake
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	P-103A — P-103A centrifugal pump — E 174'-0" N -18'-6" EL 100'-9"		
Equipment	P-103A	**Lines**	3"-P300-P-1015
Isometrics	ISO-P-1015 (owner's drawings, attached)		
Permits	General work permit		

Reason

Weeping joint at the control valve station.

Scope of work

1. Isolate per the isolation list, drain and vent 3"-P300-P-1015.
2. Break the joint at E 68'-6" N 17'-8-13/16" EL 125'-0" between VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300 and VALVE, GATE, WEDGE, OS&Y 3" P300.
3. Inspect both flange faces (SW-NDE-5).
4. Fit a new gasket and new studs; tighten to the stated preload in three passes, cross pattern (SW-WO-5).
5. Remove blinds; leak test on start-up (SW-TEST-2).

Affected items: 3"-P300-P-1015/B12, 3"-P300-P-1015/B13, 3"-P300-P-1015/B10

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1015	3"-P300-P-1015	3"-P300-P-1015	6	1 error(s), 9 warning(s) — see below

ISO-P-1015 (PDF): ../isometrics-imported/ISO-P-1015.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1015	1	error	sheet 1: CONT'D ON ISO-P-1016 prints E 174'-2 1/2" N -17'-6" EL 107'-6" for E 174'-2 1/2" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 3"-P300-P-1015 meets 3"-P300-P-1016 (E 174'-2 1/2" N -16'-6" EL 107'-6")
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W16 (E 68'-6" N -5'-0" EL 124'-7", sheet 5) has no label
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W17 (E 68'-11" N -5'-0" EL 125'-0", sheet 5) has no label
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W19 (E 69'-6" N -4'-7" EL 125'-0", sheet 5) has no label
ISO-P-1015	5	warning	weld 3"-P300-P-1015/W20 (E 69'-6" N -4'-2 1/2" EL 125'-0", sheet 5) has no label
ISO-P-1015	6	warning	W22 (3"-P300-P-1015/W22, sheet 6) is labelled again on sheet 5; it is 149.8 in from the nearest break
ISO-P-1015	6	warning	W23 (3"-P300-P-1015/W23, sheet 6) is labelled again on sheet 5; it is 159.8 in from the nearest break
ISO-P-1015	6	warning	weld 3"-P300-P-1015/W26 (E 68'-6" N 14'-11 9/16" EL 125'-0", sheet 6) has no label
ISO-P-1015	5	warning	branch weld 3/4"-P300-I-7015/W1 (E 68'-8 1/2" N -5'-0" EL 124'-0", sheet 5) has no label
ISO-P-1015	5	warning	branch weld 3/4"-P300-SP-7026/W1 (E 69'-3 1/2" N -4'-0" EL 125'-0", sheet 5) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-103A	equipment	P-103A centrifugal pump		E 174'-0" N -18'-6" EL 100'-9"
3"-P300-P-1015/B12	item	GASKET 3" P300 COMPRESSED SHEET	3"-P300-P-1015	E 68'-6" N 17'-8-13/16" EL 125'-0"
3"-P300-P-1015/B13	item	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300	3"-P300-P-1015	E 68'-6" N 17'-1-5/8" EL 125'-0"
3"-P300-P-1015/B10	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 18'-3-11/16" EL 125'-0"
3"-P300-P-1015/B16	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 15'-11-9/16" EL 125'-0"
3"-P300-P-1015/B2	item	VALVE, GATE, WEDGE, OS&Y 3" P300	3"-P300-P-1015	E 68'-6" N 21'-9-3/4" EL 125'-0"
3"-P300-P-1015/B5	item	FLANGE WELD-NECK 3" P300 RAISED FACE	3"-P300-P-1015	E 68'-6" N 21'-1/4" EL 125'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	3"-P300-P-1015/B16	3"-P300-P-1015	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1015/B16)	E 68'-6" N 15'-11-9/16" EL 125'-0"		
2	blind	3"-P300-P-1015/B13	3"-P300-P-1015	Fit a line blind between 3"-P300-P-1015/B16 and VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300 (3"-P300-P-1015/B13), work side	E 68'-6" N 17'-8-3/4" EL 125'-0"		
3	valve	3"-P300-P-1015/B2	3"-P300-P-1015	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (3"-P300-P-1015/B2)	E 68'-6" N 21'-9-3/4" EL 125'-0"		
4	blind	3"-P300-P-1015/B5	3"-P300-P-1015	Fit a line blind between 3"-P300-P-1015/B2 and FLANGE WELD-NECK 3" P300 RAISED FACE (3"-P300-P-1015/B5), work side	E 68'-6" N 21'-2-13/16" EL 125'-0"		

Parts

Description	Qty	Replaces
GASKET 3" P300 COMPRESSED SHEET	1	3"-P300-P-1015/B12
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	3"-P300-P-1015/B14

Inspections

Id	Inspection	Criteria	Basis
WO-400-04-I1	Flange faces before the new gasket goes in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
3"-P300-P-1015/B12	4 × 0.875" studs (3"-P300-P-1015/B14)	27,059	3	

Test — Service leak test

Description	The remade joint checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

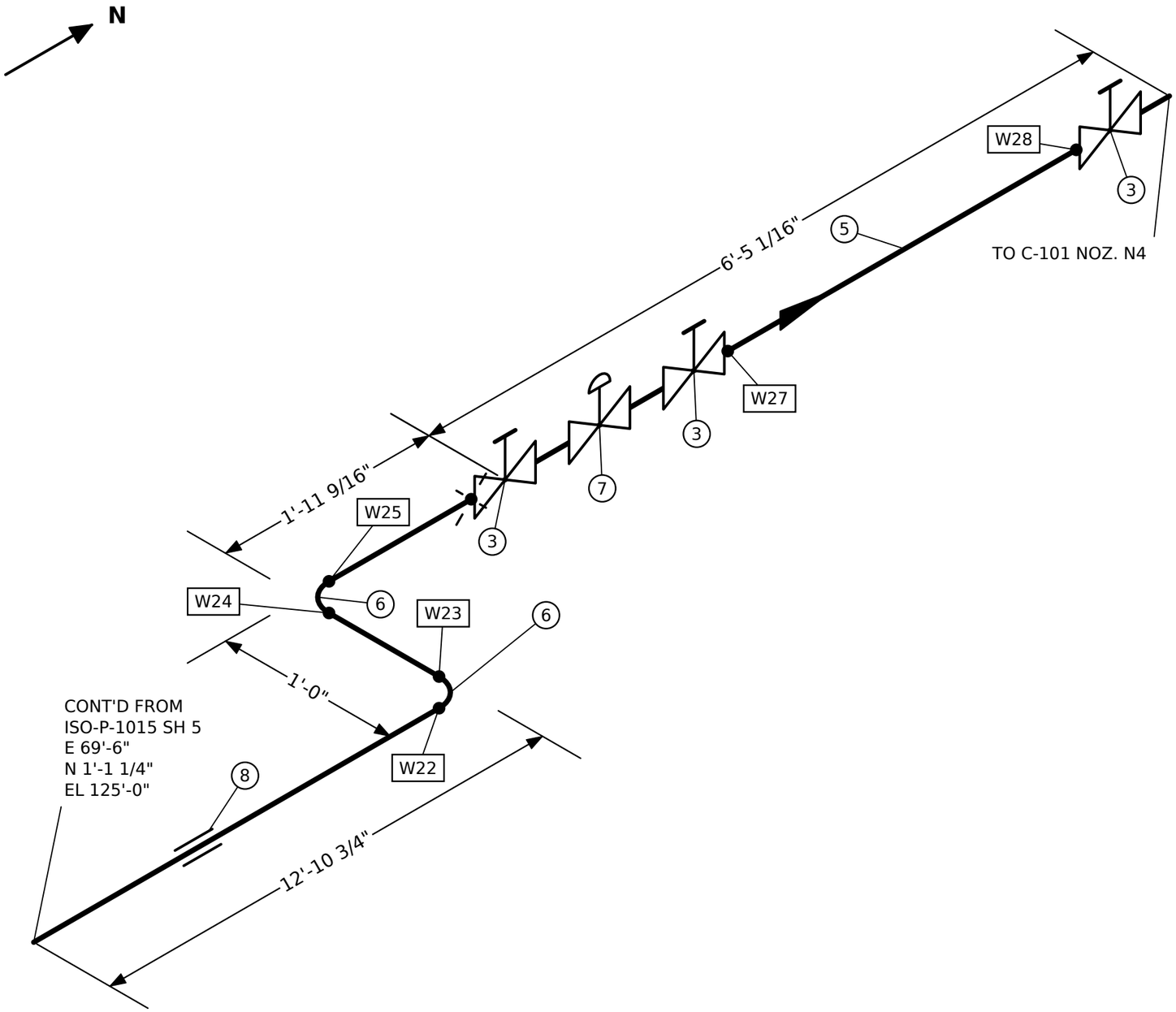
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Joint broken; both flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New gasket fitted; studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated	Bolt-up record	W	W	—				
4	Service leak test on start-up	SW-TEST-2	No leak	Leak test record	W	W	—				
5	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
6	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
3"-P300-P-1015/B12	affected	ISO-P-1015	6	
3"-P300-P-1015/B13	affected	ISO-P-1015	6	
3"-P300-P-1015/B10	affected	ISO-P-1015	6	
3"-P300-P-1015/B16	isolation: valve	ISO-P-1015	6	
3"-P300-P-1015/B13	isolation: blind	ISO-P-1015	6	
3"-P300-P-1015/B2	isolation: valve	ISO-P-1015	6	
3"-P300-P-1015/B5	isolation: blind	ISO-P-1015	6	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	6	3"	GASKET 3" P300 COMPRESSED SHEET
2	6	3"	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300
3	3	3"	VALVE, GATE, WEDGE, OS&Y 3" P300
4	3	3"	FLANGE WELD-NECK 3" P300 RAISED FACE
5	14'-8 5/16"	3"	PIPE 3" P300 BEVELLED ENDS
6	2	3"	ELBOW 90° R = 5/3 OD 3" P300
7	1	3"	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 3" P300
8	1	3"	HANGER FOR 3" LINE (HANGER-3x93.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. PAINT PER PROJECT SPECIFICATION

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3"-P300-P-1015	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1015	6 OF 6	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

Notices

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